Beyond neural predictivity: Scale tolerance as a mechanistic test of ventral-stream models

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Abstract

How the primate ventral visual stream builds invariant object representations remains a central problem. Goal-driven, task-optimized deep networks achieve high recognition accuracy and fit ventral-stream responses better than earlier mechanistic models, but predictive alignment alone does not specify mechanism. We use tolerance to retinal size as a mechanistic probe. Using large-scale object recognition to train and performance-match models, we treat this regime as a way to learn generic size-tolerant circuitry. After training, the HMAX-family model generalizes size tolerance to unfamiliar object identities, whereas performance-matched convolutional networks show tolerance that weakens for unfamiliar objects at novel sizes. The same dissociation appears in human cross-size discrimination after minimal exposure and in macaque inferotemporal recordings: neural mappings fit at one size generalize across sizes for the pooling model but not for standard convolutional networks. Intermediate representations capture scale-normalized curvature coding reported in macaque V4. These results show that conventional benchmarks can obscure distinct computational strategies.

Introduction

A central goal of systems neuroscience is to explain how the primate ventral visual stream transforms retinal inputs into representations that support rapid and invariant object recognition. Classic work by Hubel and Wiesel established a foundational

computational motif for early vision: receptive fields increase in complexity and tolerance through alternating stages of selectivity and pooling. In primary visual cortex (V1), simple cells exhibit orientation selectivity at specific retinal positions, whereas complex cells pool over similarly tuned inputs to achieve tolerance to position and phase [1]. This coupled progression of selectivity and invariance has since served as a conceptual blueprint for hierarchical accounts of object recognition in the ventral stream.

Anatomical and physiological studies later reinforced this view by demonstrating that ventral-stream areas form a serially organized hierarchy (V1, V2, V4, inferotemporal cortex; IT) [2], accompanied by increasing complexity of preferred stimuli and increasing tolerance to identity-preserving transformations such as changes in position and scale [3, 4]. These findings motivated a class of hierarchical computational models that instantiate repeated alternations of feature selectivity and pooling to build invariant representations. Fukushima's Neocognitron provided an early computational realization of this principle through cascaded layers of simple- and complex-like units [5], and subsequent theoretical work proposed that invariance could be learned from natural transformation sequences via temporal association mechanisms [6, 7].

A particularly strong empirical motivation for quantitative models of tolerance-building came from the paperclip experiments of Logothetis and colleagues [3]. Monkeys trained to recognize novel three-dimensional objects across limited rotations in depth exhibited IT neurons that were selective for object identity and view, yet tolerant to changes in retinal position and scale. Critically, substantial scale tolerance was observed despite fixed viewing distance during training, implying that tolerance to two-dimensional transformations is not learned anew for each object but instead relies on generic invariant computations available prior to object-specific learning [8]. These results suggest that the ventral stream implements mechanisms that construct tolerance in a largely stimulus-independent manner.

Riesenhuber and Poggio's HMAX model was explicitly designed to account for these properties [9]. HMAX instantiates a feedforward hierarchy of alternating simple (S) and complex (C) stages, in which selectivity increases through template matching while tolerance emerges through pooling over afferent units with similar tuning. In this framework, generic invariance to position and scale is built gradually through cascaded pooling operations across layers, and higher-level units then combine these invariant features to achieve object- and view-specific selectivity. A key prediction—that complex cells implement a max-like pooling operation—was later supported by intracellular recordings showing that complex-cell responses are well described by max-like nonlinearities over simple-cell inputs [10]. Subsequent work extended HMAX into quantitatively specified ventral-stream models and demonstrated that feedforward implementations could account for both neural responses and human performance in rapid visual categorization tasks [11–13].

The deep learning revolution in vision enabled dramatic gains in object recognition performance and spurred architectural innovations such as VGG, Inception, and residual connections [14–18]. For neuroscience, these advances helped establish a goal-driven, task-optimized modeling strategy: optimize a network to solve an ethologically relevant recognition task and then test its internal representations against

neural measurements. Seminal work showed that such task-optimized networks could predict responses in macaque IT and V4, often outperforming earlier biologically inspired models [19, 20], and this relationship was later formalized in standardized benchmarks such as Brain-Score [21].

A substantial body of work has explicitly examined invariance properties in deep neural networks. Early studies introduced quantitative metrics to measure how learned representations change under identity-preserving transformations, showing that invariance tends to increase with depth but remains modest and strongly dependent on training objectives and data [22]. Subsequent work demonstrated that convolution and pooling alone do not guarantee broad translation or scale invariance in practice: performance often degrades outside the range of transformations encountered during training, although targeted augmentation can induce tolerance over a limited range [23]. Analyses of modern ImageNet-trained networks similarly show that scale invariance increases with depth but remains incomplete, with intermediate layers retaining substantial scale sensitivity [24, 25]. Together, these findings indicate that contemporary CNNs can exhibit nontrivial invariance, but that its extent and form depend critically on architecture, training regime, and the transformations considered.

These developments raise a fundamental question for neuroscience: when a model achieves high benchmark performance and strong neural predictivity, what aspects of its internal computation are genuinely explanatory of ventral-stream processing? Benchmark accuracy and neural predictivity do not uniquely determine mechanism. Scale tolerance provides an informative lens on this problem. Humans can generalize across substantial changes in retinal size after minimal exposure [26], whereas modern convolutional networks often rely on extensive scale augmentation during training and exhibit sharp performance degradation outside the augmentation range [27]. At the neural level, mid-level ventral-stream representations exhibit transformation-tolerant codes that remain stable across changes in retinal size, including curvature-based invariances in area V4 [28].

Here we test whether scale tolerance can serve as a mechanistic probe of ventral-stream models at modern performance levels. We systematically modernize the HMAX framework while preserving its tolerance-building organization and compare it to standard convolutional networks across object recognition, behavioral generalization, and neural alignment.

Results

We asked whether models that achieve comparable task performance and conventional neural predictivity nevertheless differ in how they construct invariant visual representations. We used large-scale object recognition both to train these models and to match their overall performance, then probed whether the resulting representations generalize scale tolerance to unfamiliar objects. We compared modernized HMAX-family models and standard convolutional networks across four analyses: (i) large-scale recognition performance, (ii) generalization across retinal size when test objects differ from those used to train the model, (iii) human cross-size discrimination

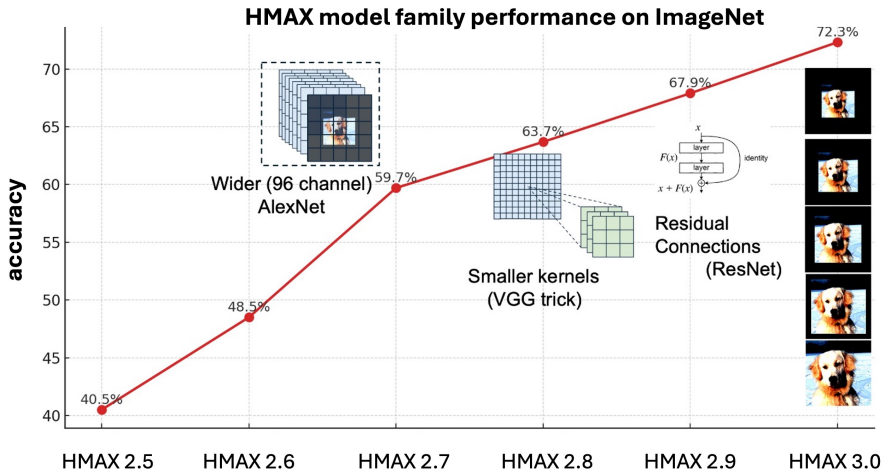

Fig. 1 Architectural progression of HMAX variants and ImageNet performance.

after minimal exposure, and (iv) neural alignment and physiological signatures along the primate ventral stream.

Modernizing a ventral-stream architecture preserves object recognition performance

A common concern regarding biologically grounded models is that architectural commitments may limit performance on large-scale recognition tasks. To test this directly, we progressively modernized the HMAX architecture to operate in a contemporary performance regime while preserving its core tolerance-building organization. These updates included replacing fixed feature extractors with learnable convolutional filters, increasing depth via residual connections, and enabling end-to-end optimization.

Across this progression, each modernization step yielded systematic gains in ImageNet top-1 accuracy (Figure 1). The final model achieved performance comparable to widely used convolutional architectures in this regime. Importantly, these gains were obtained without abandoning explicit multi-scale processing or hierarchical tolerance-building, establishing that ventral-stream–inspired model families are not intrinsically confined to low-performance regimes. This provides a controlled basis for comparing invariance at similar levels of task accuracy.

Models differ in how they generalize across retinal scale

We next examined whether performance-matched models differ in their ability to generalize across changes in retinal object size. To isolate intrinsic scale tolerance from effects of augmentation, models were trained at a single reference scale and evaluated across a range of test scales spanning approximately one octave.

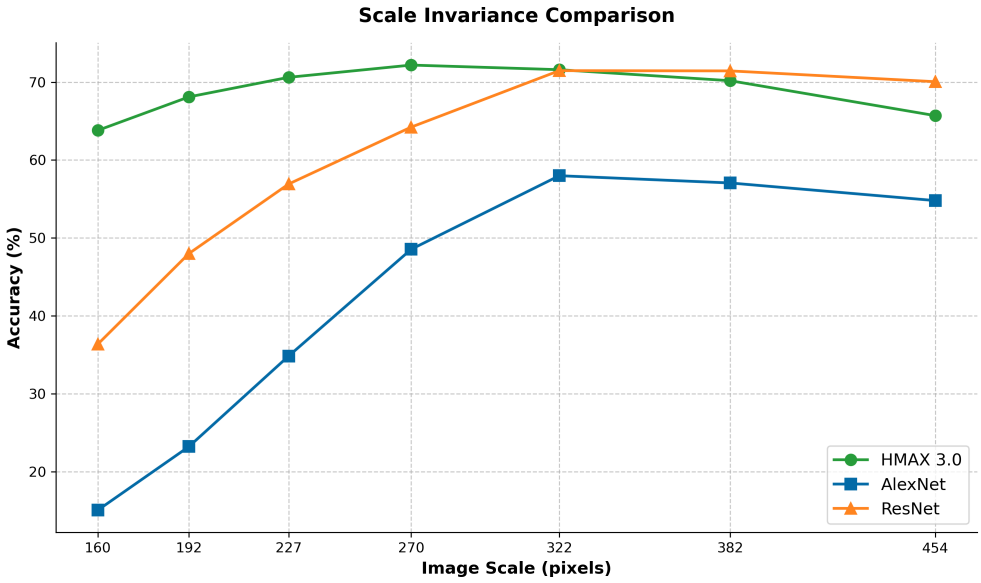

Fig. 2 ImageNet scale generalization across seven test scales.

Under this off-reference evaluation, standard convolutional networks showed pronounced degradation as test scale deviated from the training scale (Figure 2). In contrast, modernized HMAX-family models maintained stable performance across the same range. This difference persisted despite comparable accuracy at the reference scale, indicating that it is not explained by overall task difficulty or underfitting.

Consistent with prior work, these results do not imply that convolutional networks lack scale tolerance altogether, but rather that such tolerance is typically restricted to the range of transformations represented in the training data and does not generalize reliably beyond that range [23, 24].

Human-like one-shot generalization distinguishes model families

To assess whether the differences observed above extend to regimes relevant for perception, we evaluated models on a one-shot discrimination task using unfamiliar Hangul characters. This paradigm probes generalization of identity across large size changes after minimal exposure, a hallmark of human visual recognition.

Modernized HMAX-family models matched human performance under out-of-distribution scale conditions, whereas standard convolutional networks showed marked declines despite strong in-distribution performance (Figure 3). This dissociation mirrors the ImageNet scale generalization results and demonstrates that differences in scale tolerance manifest not only in large-scale classification benchmarks but also in controlled behavioral paradigms that approximate human perceptual demands.

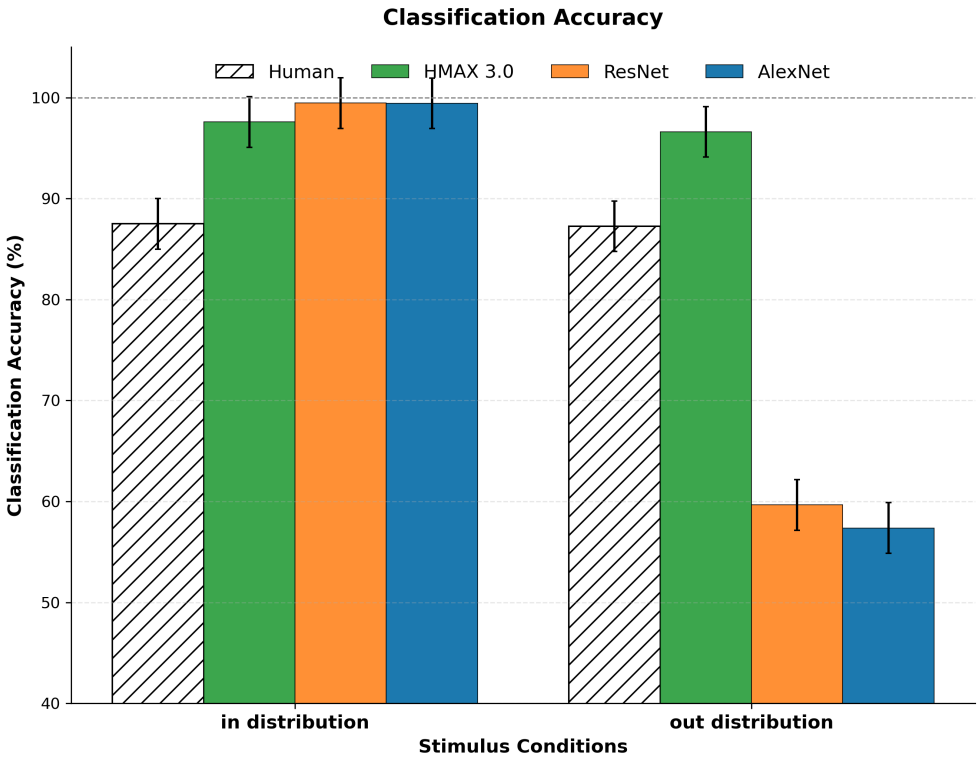

Fig. 3 One-shot Hangul discrimination across scale.

Neural predictivity diverges under scale perturbation

We next asked whether differences in scale tolerance are reflected in alignment with neural responses in inferotemporal cortex. Using the Brain-Score framework, we first confirmed that modernized HMAX models and standard convolutional networks can achieve comparable neural predictivity under conventional evaluation protocols.

We then applied a *scale-split* analysis as a diagnostic probe of generalization beyond the regime used to fit neural mappings (rather than as a new benchmark): mappings were fit at a reference scale and evaluated on stimuli at unseen scales. Under this procedure, a clear divergence emerged (Figure 4). Neural predictivity of standard convolutional networks declined substantially under scale shifts, whereas modernized HMAX models preserved alignment across scale. Notably, this divergence was not apparent under standard evaluation, indicating that single-operating-point predictivity can obscure differences in how models represent identity-preserving transformations.

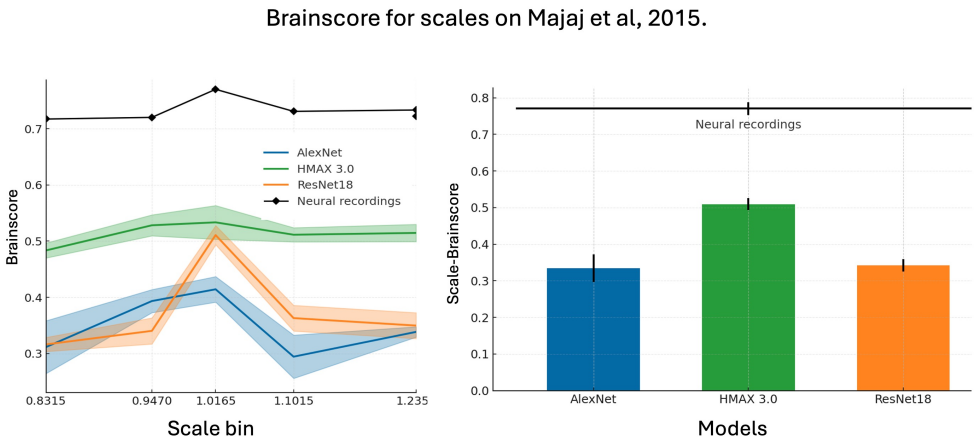

Fig. 4 Brain-Score performance under scale variation.

Model	Layer	Invariance Score	MC χ^2	p
HMAX 3.0	C2	0.080	8.58	0.21
AlexNet	Best	0.609	26.76	0.0004
ResNet-18	Best	0.754	31.82	< 0.001

Table 1 Replication of scale-normalized curvature tuning in macaque V4.

Intermediate representations capture a physiological signature of V4 invariance

Finally, we tested whether architectural differences in scale tolerance are reflected at intermediate stages of processing. We replicated a classic electrophysiological paradigm demonstrating that neurons in macaque area V4 encode contour curvature in a scale-normalized manner, maintaining tuning stability across changes in retinal size [28].

Intermediate layers of the modernized HMAX hierarchy reproduced normalized curvature tuning consistent with macaque V4, whereas no layer in standard convolutional networks exhibited comparable stability, even when selecting the best-matching layer post hoc (Table 1). This result links differences observed at the behavioral and benchmark levels to differences in how intermediate representations encode shape information across scale.

Methods

Reproducibility and implementation details.

For complete reproducibility, we provide the full set of architectural specifications, training hyperparameters, augmentation policies, and evaluation protocols in Supplementary Section S1.1. **[FILL: Fill Supplementary Section S1.1 with the exact settings used for all results reported in this manuscript.]**

Our goal was to test how different modeling approaches construct invariant visual representations when models are scaled to contemporary performance regimes. To this end, we compared modernized HMAX-family architectures to standard convolutional neural networks under matched evaluation protocols that probe object recognition performance, out-of-distribution generalization across scale, and alignment with neural and physiological data from the primate ventral stream.

Model architectures

We evaluated a family of progressively modernized HMAX architectures alongside standard convolutional baselines (e.g., AlexNet- and ResNet-style models). The HMAX-family models preserve the core organizational principles of the original framework: explicit multi-scale processing, alternating stages of feature selectivity and pooling, and progressive construction of tolerance across a hierarchical feedforward pathway.

To operate in a modern performance regime, HMAX architectures were updated with learnable convolutional filters, residual connections, and end-to-end optimization. Fixed Gabor filters were replaced with trainable convolutional kernels, and deeper hierarchies were enabled via residual connections while preserving the simple/complex (S/C) structure. Explicit scale channels were implemented using dense sampling of image scale, and scale information was integrated hierarchically through differentiable pooling operations rather than hard max selection. These design choices allow the model to remain trainable at scale while retaining a ventral-stream–like organization.

Standard convolutional networks served as performance-matched baselines. All models were trained and evaluated using identical datasets and task definitions unless otherwise noted.

Training procedures

Models were trained using supervised learning for object categorization, with cross-entropy loss optimized via stochastic gradient descent–based methods. To isolate architectural contributions to scale tolerance, HMAX-family models were trained without extensive scale jitter, whereas baseline convolutional networks were trained using conventional augmentation practices unless explicitly stated.

For HMAX architectures, an additional hierarchical scale-alignment objective was applied during training to encourage consistency of representations across adjacent scales throughout the hierarchy. This regularization promotes gradual tolerance building in a manner consistent with ventral-stream processing. Full architectural

specifications, optimization parameters, and training schedules are reported in the Supplementary Materials.

Scale generalization protocols

To assess intrinsic scale tolerance, we adopted an off-reference evaluation paradigm. Models were trained at a single reference scale and tested across a range of retinal scales spanning approximately one octave. Performance was quantified both at the reference scale and on held-out scales to distinguish in-distribution accuracy from true extrapolative generalization.

This protocol was applied consistently across datasets, including ImageNet-scale natural images and controlled benchmark stimuli. By decoupling training and test scales, this evaluation isolates mechanisms of scale tolerance from effects driven by data augmentation.

Behavioral evaluation: one-shot generalization

To probe human-like generalization, we evaluated models on a one-shot discrimination task using unfamiliar Hangul characters, following established psychophysical procedures. Models were exposed to a single example of a target character and required to discriminate it from visually similar distractors under varying scale conditions.

Model decisions were based on similarity in feature representations extracted from the penultimate layer. Performance was assessed separately for in-distribution and out-of-distribution scale conditions and compared to behavioral data from human observers. Sensitivity was quantified using signal detection metrics where appropriate.

Neural evaluation: Brain-Score and scale-split analysis

Neural alignment was assessed using the Brain-Score framework, which quantifies the correspondence between model representations and neural responses recorded from macaque inferotemporal cortex. For each model, linear mappings from model activations to neural responses were learned using standard cross-validated procedures, and predictivity was measured as noise-ceiling–normalized correlation.

To probe generalization beyond the fitting regime, we implemented a scale-split evaluation. Linear mappings were fit using stimuli presented at a reference scale and evaluated on neural responses to stimuli at unseen scales. This procedure tests whether neural predictivity reflects robust, scale-tolerant representations or scale-specific correlations.

Physiological validation in area V4

To test whether intermediate model representations reproduce known physiological signatures of ventral-stream processing, we replicated a classic paradigm examining scale-normalized curvature tuning in macaque area V4 [28]. Model units were probed with the same class of parametric shape stimuli across multiple retinal sizes.

For each responsive unit, tuning curves were computed as a function of curvature at each scale, and invariance was quantified by measuring the stability of tuning

preferences across scale. Model-derived distributions were compared directly to published V4 data using identical metrics and statistical tests, enabling a mechanistic comparison between artificial and biological representations.

Discussion

The present study revisits a biologically grounded model of the ventral visual stream to test which computational principles remain explanatory when models are scaled to contemporary performance regimes. By systematically modernizing the HMAX framework while preserving its core organizational commitments, we show that different modeling approaches can achieve similar levels of task performance and neural predictivity while diverging in how invariant visual representations are constructed. Together, these results clarify how invariant representations depend on the form of computation rather than task optimization alone.

Architectural differences and the form of invariance

A central finding of this work is that performance-matched models can differ markedly in the *form* of invariance they implement. Modern convolutional networks typically achieve scale tolerance through extensive data augmentation, yielding high in-distribution accuracy but brittle extrapolation beyond the training range [26, 27]. In contrast, modernized HMAX-family models exhibit robust generalization across scale without reliance on large numbers of scale-augmented examples. This distinction is evident across natural-image benchmarks, one-shot behavioral tasks, and neural evaluations, indicating that differences in tolerance-building mechanisms lead to qualitative differences in representation that are not captured by aggregate performance metrics.

Importantly, this conclusion does not contradict evidence that CNNs can support substantial invariance under appropriate training conditions. Psychophysical comparisons show that both humans and CNNs can exhibit robust online translation tolerance following extreme displacements, provided that the networks are trained with suitable data and architectural choices [29]. However, such invariance is typically acquired through exposure to specific transformations and does not appear to generalize broadly across untrained regimes, nor does it necessarily align with the invariance profiles observed in human perception [25]. The present results extend this literature by showing that models with similar task accuracy and neural predictivity can nevertheless differ in how invariance is constructed and generalized.

These results support the view that invariance in the ventral stream is not merely a statistical consequence of exposure to transformed exemplars, but reflects computational structure that builds tolerance progressively across hierarchical stages [3, 4]. Explicit multi-scale processing and hierarchical pooling provide one plausible mechanism for constructing such representations, consistent with longstanding theories of ventral-stream organization.

Implications for neural predictivity benchmarks

Our findings refine the interpretation of neural predictivity benchmarks such as Brain-Score [21]. While modern deep networks often achieve strong correspondence with neural responses under standard evaluation protocols [19, 20], this alignment does not uniquely constrain underlying computation. By introducing a scale-split evaluation, we show that models with similar Brain-Score values can diverge substantially when tested outside the regime used to fit neural mappings. In this setting, models that preserve scale tolerance maintain neural predictivity, whereas standard convolutional networks do not.

These results do not undermine the utility of neural predictivity benchmarks; rather, they highlight the importance of probing generalization beyond the fitting regime. Neural predictivity provides an important constraint, but it is insufficient on its own to adjudicate between competing mechanistic accounts of visual representation. Incorporating targeted perturbations that probe specific invariances may therefore be critical for evaluating candidate models as theories of cortical computation.

Prediction, explanation, and levels of analysis

The dissociations observed here bear on a long-standing distinction between predictive adequacy and mechanistic explanation. Models optimized for task performance can achieve strong neural alignment while implementing qualitatively different computational strategies. This separation echoes classical distinctions between levels of analysis in cognitive science, in which multiple algorithmic or mechanistic realizations can satisfy the same computational goal.

Recent advances in deep learning raise the possibility that increasingly flexible and scalable models may eventually discover representations resembling those shaped by biological evolution. However, the present results show that models with comparable predictive accuracy and neural predictivity can diverge systematically in how invariance is constructed, and that these differences become apparent only under targeted tests that probe generalization beyond the training distribution. Thus, even if data-driven optimization can recover brain-like behavior in some regimes, mechanistic convergence cannot be assumed.

From this perspective, biologically grounded models should not be viewed as architectural prescriptions, but as experimental instruments that enable controlled tests of competing mechanistic hypotheses. Scale tolerance provides one such test case. More broadly, these findings underscore the need to complement predictive benchmarks with hypothesis-driven probes if computational models are to move from prediction toward explanation [30].

Behavioral and physiological convergence

The convergence of behavioral and physiological results strengthens the interpretation of scale tolerance as a biologically meaningful diagnostic. Modernized HMAX models reproduce human-like generalization in one-shot discrimination tasks that require extrapolation across scale [26], a regime in which standard convolutional networks fail

despite comparable in-distribution accuracy. At the physiological level, intermediate model representations reproduce normalized curvature tuning in macaque area V4 [28], a hallmark signature of scale-invariant shape encoding. The absence of this signature in contemporary convolutional architectures underscores the value of mechanistic tests that go beyond correlational alignment.

Taken together, these findings suggest that scale tolerance provides a particularly informative window into ventral-stream computation. More broadly, they illustrate how targeted probes of invariant coding can reveal differences in representational structure that remain hidden under standard task and neural benchmarks.

Limitations and future directions

Several limitations of the present work point to important directions for future research. First, the models studied here are purely feedforward, whereas biological vision involves extensive recurrent and feedback interactions that contribute to attention, contextual modulation, and temporal integration [4]. Incorporating such dynamics may further refine invariant representations and improve alignment with neural responses. Second, while scale tolerance provides a clear and tractable test case, other forms of invariance—such as tolerance to viewpoint, illumination, and occlusion—remain to be examined within the same framework. Finally, the present study focuses on visual object recognition; extending these approaches to multimodal and task-dependent settings will be necessary to assess the generality of the proposed principles.

Conclusions

By revisiting a biologically grounded ventral-stream model under modern performance constraints, this work demonstrates that predictive success and neural alignment do not uniquely specify underlying computation. The results show that invariant visual representations depend on how tolerance is constructed within a model, not solely on task optimization. More generally, they underscore the importance of targeted mechanistic tests in evaluating computational models as explanations of cortical processing.

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Supplementary Information

Beyond neural predictivity: Scale tolerance as a mechanistic test of ventral-stream models

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S1 Supplementary Methods

S1.1 Reproducibility checklist (to be completed)

This section is intentionally written as a fillable checklist. Please replace each **[FILL: ...]** and **[VERIFY: ...]** item with the exact settings used in the experiments, or delete the item if it does not apply.

Code, environment, and hardware

- Code repository URL (private is fine for now): **[FILL: repo URL]**
- Exact commit hash for experiments reported in this paper: **[FILL: git commit]**
- Framework + version (e.g., PyTorch 2.3.1 / CUDA 12.1): **[FILL: framework + versions]**
- GPU(s) used (model, count), CPU, RAM: **[FILL: hardware]**
- Total training time per model (approx.): **[FILL: time/model]**
- Random seeds: **[FILL: seed(s)]** **[VERIFY: state whether results are from 1 run or mean across multiple seeds]**

Datasets and preprocessing

- ImageNet version (ILSVRC2012?) and exact splits: **[FILL: dataset + split]**
- Brain-Score benchmark(s) used (exact benchmark names/IDs): **[FILL: benchmark names]**
- Hangul stimuli source files / generation script: **[FILL: location or citation]**
- Pasupathy/El-Shamayleh stimuli generation or download details: **[FILL: location or citation]**

Input pipeline details (ImageNet).

- Training preprocessing (resize/crop/normalization): **[FILL: exact pipeline]**
- Test preprocessing at the *reference scale*: **[FILL: exact pipeline]**
- Scale evaluation preprocessing (how the seven scales are applied relative to crop/resize): **[FILL: exact pipeline]**
- **[VERIFY: clarify how "training on a single reference scale (227×227)" relates to the base input size used elsewhere (e.g., 322×322)]**

Model architectures (exact specification)

HMAX 3.0.

- Full list of image pyramid scales used at train and test time (e.g., $\Sigma = \{\sigma_1, \ldots, \sigma_M\}$): **[FILL: scale list]**
- Channel widths per block/layer (S1/S2/S2b/S3 etc.): **[FILL: channels]**
- Kernel sizes, strides, padding, activation, normalization: **[FILL: layer specs]**
- Definition of the scale scorer k_φ (architecture, sharing across layers): **[FILL: scorer]**
- Definition of the "feature adapter" used before adjacent-scale pooling: **[FILL: adapter]**

- Where (which layers) pairwise scale pooling is applied, and when global pooling is applied: **[FILL: pooling placement]**

Baselines (AlexNet / VGG-16 / ResNet-18).

- Source implementation (torchvision? custom?) and any modifications: **[FILL: source + mods]**
- Parameter counts reported in Table S2: **[VERIFY: confirm these values and how they were computed]**

Training hyperparameters (must be explicit)

Objective and regularization.

- Classification loss (cross-entropy etc.): **[FILL: loss]**
- Alignment loss weights λ_l for $l \in \{C1, C2, \text{Out}\}$: **[FILL: $\lambda_{C1}, \lambda_{C2}, \lambda_{Out}$]**
- Any additional regularizers (weight decay, dropout, label smoothing, mixup/cutmix, EMA): **[FILL: regularizers]**

Optimizer and schedule.

- Optimizer (SGD/AdamW/etc) + settings (momentum/betas): **[FILL: optimizer]**
- Batch size, epochs, gradient clipping (if any): **[FILL: batch/epochs/clipping]**
- Learning rate, warmup, schedule (cosine/step), minimum LR: **[FILL: LR schedule]**
- Weight decay: **[FILL: wd]**

Augmentation policy (critical for scale-invariance claims).

- Scale jitter range used for *each* model (HMAX and baselines): **[FILL: scale-jitter per model]**
- Other augmentations (color jitter, flips, RandAugment, etc.): **[FILL: augmentations]**
- **[VERIFY: explicitly state whether HMAX 3.0 is trained without multi-scale jitter, and if not, quantify the exact augmentation window]**

Evaluation protocols (exact implementation)

ImageNet scale protocol.

- Exact seven evaluation scales used and their interpretation: **[FILL: scales + meaning]**
- Off-reference score definition (confirm Equation used): **[FILL: equation ref or text]**
- Number of evaluation images and whether results are single-crop or multi-crop: **[FILL: eval details]**

Hangul one-shot protocol.

- Pixel-to-visual-angle mapping (e.g., 26 px = 1°): **[VERIFY: confirm mapping]**

- Feature layer used (penultimate?), feature normalization (if any): **[FILL: layer + processing]**
- Pearson-correlation threshold selection (fixed? tuned on validation? per model?): **[FILL: threshold procedure]**
- d′ computation details (hit/false-alarm definitions) + citations: **[FILL: details]** **[CITE]**

Brain-Score protocol.

- Exact benchmark(s) and stimulus sets used (e.g., "MajajHong2015" variants): **[FILL: benchmark IDs]**
- PLS parameters: number of components, regularization, cross-validation folds: **[FILL: PLS details]**
- Which model layers were evaluated and how the "best layer" was selected: **[FILL: layer set + selection]**
- Scale-split protocol: bin definitions, reference bin, train/test split: **[FILL: scale-split details]**

Pasupathy / V4 normalized-curvature replication.

- RF size computation method and exact layer-wise RF sizes used: **[FILL: RF sizes]**
- Z-score neuron selection threshold (e.g., $Z > 2$): **[VERIFY: confirm threshold]**
- Binning used for χ^2 test and Monte Carlo procedure details: **[FILL: bins + MC details]**

Reporting and statistics

- Number of independent runs per model and variability reporting (mean±SD or CI): **[FILL: runs + variability]**
- Statistical tests used (where), multiple-comparison correction (if any): **[FILL: stats]**
- What constitutes "state-of-the-art" in each benchmark (with citations): **[FILL: SOTA comparisons]** **[CITE]**

S1.2 HMAX 3.0 architectural details (ported from student draft)

HMAX 3.0 is a backpropagation-enabled, scale-invariant architecture that builds on the principles of the classical HMAX model by integrating modern deep learning mechanisms. The key components are summarized below.

Architectural modernization via learnable kernels.

To enhance feature expressivity, we adopt the channel expansion strategy of AlexNet and widen early layers (e.g., S1 and S2). Following VGG, large receptive field filters are decomposed into stacks of smaller kernels (e.g., sequences of 3×3 convolutions). To ensure trainability at depth, we incorporate residual skip connections within major feature extraction blocks (S1, S2, S2b, and S3).

Image pyramid and dense scale sampling.

Rather than applying filters at multiple scales, we use an image pyramid in which input images are explicitly rescaled to different sizes. Motivated by Gaussian scale-space theory, we sample scales densely at quarter-octave increments (scale factor $2^{1/4}$), enabling smooth transitions across scale space.

Adaptive pooling and a high-resolution bypass stream.

To maintain spatial coherence across different input scales, we use adaptive stride pooling in early C layers. To balance invariance with localization, we include a bypass stream (S2b) branching from C1 that preserves higher-resolution features; outputs of the deep semantic stream and bypass stream are concatenated prior to classification.

S1.3 Hierarchical learnable scale selection (ported from student draft)

A hard max over scales is the simplest way to merge a pyramid:

$$\tau(x) = \max_{\sigma \in \Sigma} F_\sigma(x), \tag{S1}$$

where $\Sigma = \{\sigma_1, \ldots, \sigma_M\}$ are sampled scales, T_σ is the rescaling operator, and $F_\sigma(x) = f_\theta\big(T_\sigma x\big)$ is the feature representation at resolution σ.

HMAX 3.0 replaces hard max pooling with a differentiable, data-driven operator. A lightweight scorer k_φ produces per-scale scores $s_\sigma(x) = k_\varphi\big(F_\sigma(x)\big)$, which are converted to weights via a softmax. The pooled descriptor is computed as a weighted sum:

$$\phi(x) = \sum_{\sigma \in \Sigma} \alpha_\sigma(x)\, F_\sigma(x). \tag{S2}$$

This mechanism is applied progressively to preserve semantic coherence: (i) *pairwise intermediate pooling* merges only adjacent scales in intermediate layers, after a learnable feature adapter aligns them into a common subspace; and (ii) *global final pooling* aggregates across remaining scales at the final integration stage.

S1.4 Scale-invariant regularization via adjacent-pair alignment (ported from student draft)

During training, the same image is processed at two adjacent scales (σ_i, σ_{i+1}). Scale consistency is enforced hierarchically throughout the network by computing an L_1 distance between corresponding feature maps:

$$\mathcal{L}_{\text{align}} = \sum_{l \in \{C1, C2, \text{Out}\}} \lambda_l \left\| \phi_l(T_{\sigma_i} x) - \phi_l(T_{\sigma_{i+1}} x) \right\|_1 . \tag{S3}$$

The complete objective combines classification and alignment:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{classification}} + \mathcal{L}_{\text{align}}. \tag{S4}$$

S2 Supplementary Experimental Protocols

S2.1 ImageNet scale generalization and off-reference scoring (ported from student draft)

Input image sizes were systematically varied using seven scales (quarter-octave steps relative to a base size of 322×322 pixels): $\{160, 192, 227, 270, 322, 382, 454\}$. To isolate intrinsic scale tolerance, models can be trained at a single reference scale (e.g., 227×227) and evaluated across all scales. The off-reference score excludes the reference scale:

$$\text{Off-Reference Score} = \frac{1}{|\mathcal{S}_{\text{off}}|} \sum_{s \in \mathcal{S}_{\text{off}}} \text{Accuracy}(s). \tag{S5}$$

S2.2 Behavioral evaluation: one-shot generalization on Hangul (ported from student draft)

We used the Hangul one-shot discrimination paradigm of Han et al. [26]. The dataset contains 27 target/distractor pairs of visually similar characters. Five scale conditions are defined by target/test visual angles (in degrees): $30°/2°$, $2°/30°$, $30°/5°$, $5°/30°$, and $30°/30°$ (in-distribution). Out-of-distribution performance averages the four cross-scale conditions.

For mapping degrees to pixels in our model stimuli, we used a conversion of 26 pixels $= 1°$ of visual angle. Two characters were classified as the same identity when the Pearson correlation between their feature vectors exceeded a threshold (threshold selection procedure should be stated explicitly: e.g., fixed a priori vs. tuned on a held-out set). Out-of-distribution (OOD) performance averages across the four cross-scale conditions ($30°/2°$, $2°/30°$, $30°/5°$, and $5°/30°$), while the $30°/30°$ condition defines in-distribution performance.

Model decisions are based on Pearson correlation between penultimate-layer feature vectors for the two characters in a pair, thresholded to classify "same" vs. "different". Sensitivity is quantified using d':

$$d' = Z(\text{Hit Rate}) - Z(\text{False Alarm Rate}),$$

where Z is the inverse standard normal CDF. *(TODO: replace placeholder citations for d', e.g., Green & Swets; Macmillan & Creelman, and ensure entries exist in* `neurips_2025.bib`*).*

S2.3 Neural evaluation: Brain-Score and scale-split analysis (ported from student draft)

We evaluated neural predictivity on macaque IT recordings in the Brain-Score framework [31**?**]. For each model layer l, a linear mapping from activations $\mathbf{X}^{(l)}$ to neural responses $\mathbf{Y}$ is learned using cross-validated partial least squares regression, and performance is reported as noise-ceiling–normalized Spearman correlation.

Specifically, Brain-Score fits a linear mapping from model activations to recorded neural responses using cross-validation (e.g., partial least squares regression), then

reports predictivity as noise-ceiling–normalized Spearman correlation. Following the student draft, the reported model score should be the *maximum* across evaluated layers (i.e., the best-aligned layer), and the scale-split protocol trains the mapping on a reference scale bin and evaluates generalization on held-out images from all bins.

To probe extrapolation across scale, we implemented a scale-split protocol in which the mapping is fit using images from a reference scale bin and evaluated on images from all scale bins; we also report an off-scale metric that excludes the reference bin:

$$\text{Brain-Score}_{\text{off-scale}} = \frac{1}{|\mathcal{B}_{\text{off}}|} \sum_{b \in \mathcal{B}_{\text{off}}} \text{Brain-Score}^{(b)}. \tag{S6}$$

S2.4 Physiological validation: scale-normalized curvature tuning in area V4 (ported from student draft)

We replicated the El-Shamayleh & Pasupathy paradigm [28] using parametric

Stimuli were generated at multiple curvature levels and presented at 8 rotations (0° to 315° in 45° steps). To test scale invariance, each stimulus was presented at four relative sizes: 0.4×, 0.6×, 0.8×, and 1.0×. For each layer, stimuli were cropped to the receptive field size of the central unit and resized to the target scale, then padded back to the full model input size to mimic foveal presentation (RF sizes computed analytically; see [**?**] in the student draft). Units were included for analysis if their peak Z-scored response exceeded $Z > 2.0$. curvature stimuli across rotations and sizes. Responsive units are selected by Z-score filtering: a unit is included if its peak response across the full stimulus set satisfies $Z > 2.0$ relative to its own response distribution. Invariance is quantified by the stability of the tuning-curve centroid across scale; per-unit slopes are aggregated as the median absolute slope.

S3 Supplementary Results

S3.1 Comprehensive performance table (ported from student draft)

Evaluation Domain	**Metric**	**HMAX 3.0**	**Best Baseline**	**Improveme**
ImageNet Classification	Top-1 Accuracy	72.3%	69.8% (ResNet18)	+2.5%
Scale Invariance (MNIST)	Off-reference accuracy	85.2%	62.1% (VGG-16)	+23.1%
One-shot Learning (Hangul)	Cross-scale accuracy	78.4%	65.2% (ResNet18)	+13.2%
Hangul d-prime Analysis	Mean d-prime	0.90	0.21 (AlexNet)	+329%
Brain-Score (Red Central IT)	Neural prediction	0.613	0.487 (AlexNet)	+25.8%
Brain-Score (George Post. IT)	Neural prediction	0.549	0.423 (VGG-16)	+29.8%
Brain-Score (George Central IT)	Neural prediction	0.536	0.298 (ResNet18)	+79.9%

Table S1 Comprehensive performance comparison across evaluation domains (ported from student draft).

Model	Parameters (M)
HMAX 3.0	128.9
AlexNet	61.1
ResNet-18	11.69

Table S2 Comparison of model parameters (ported from student draft).

S3.2 Supplementary figures (ported from student draft)

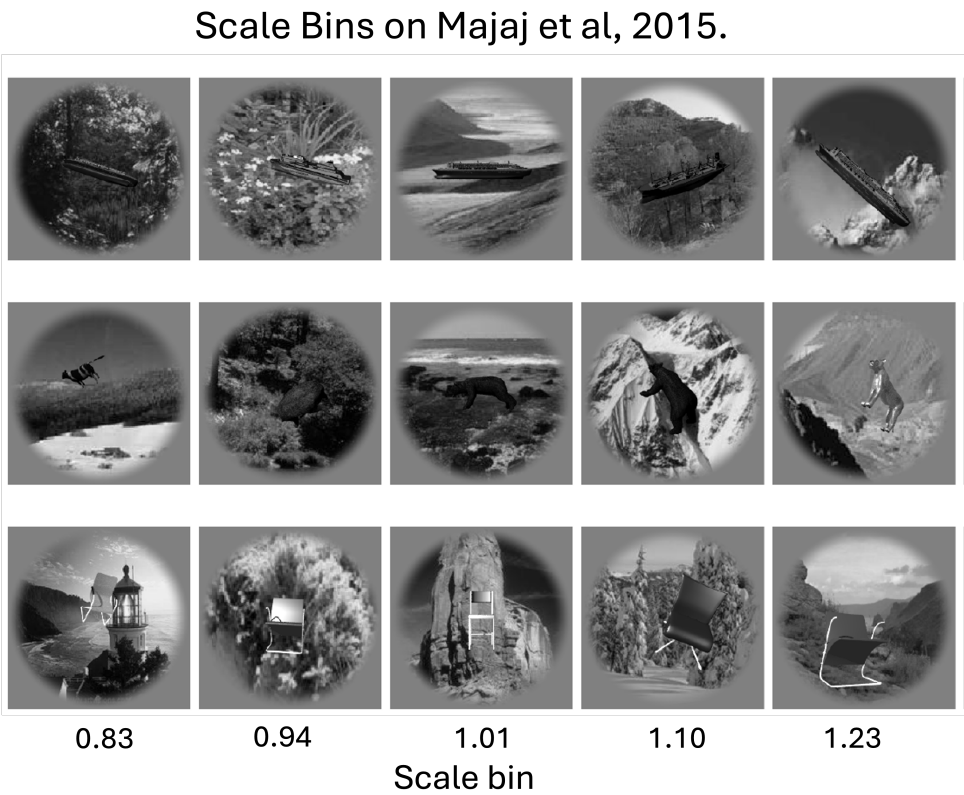

Fig. S1 Example stimuli from the Brain-Score benchmark dataset (ported from student draft).

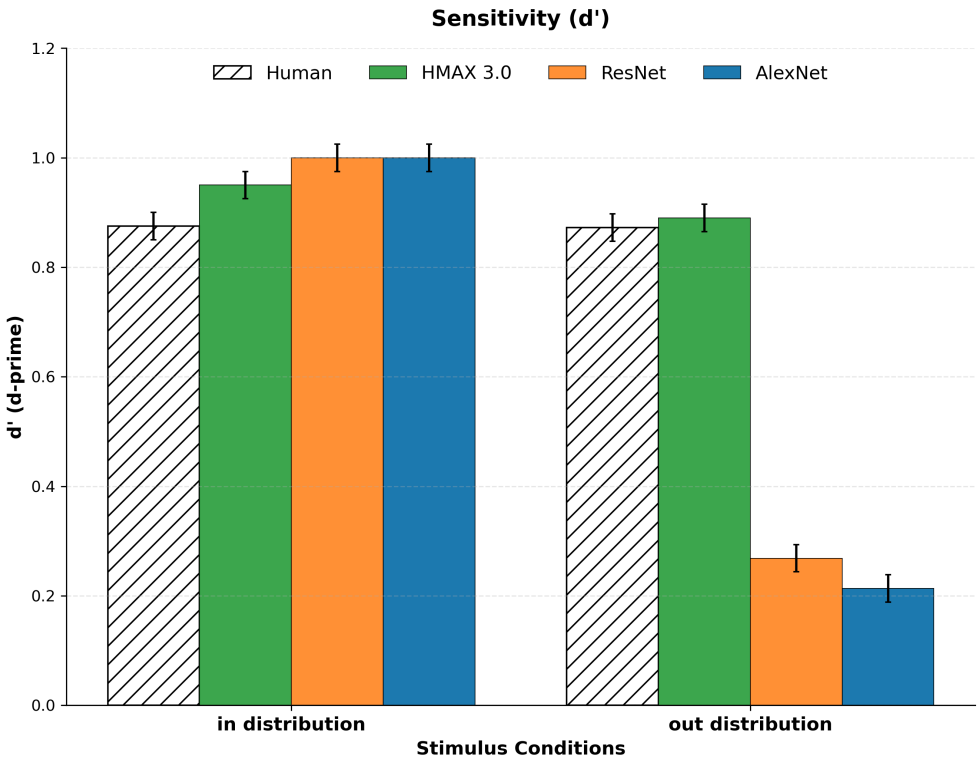

Fig. S2 Hangul one-shot learning sensitivity analysis (ported from student draft).

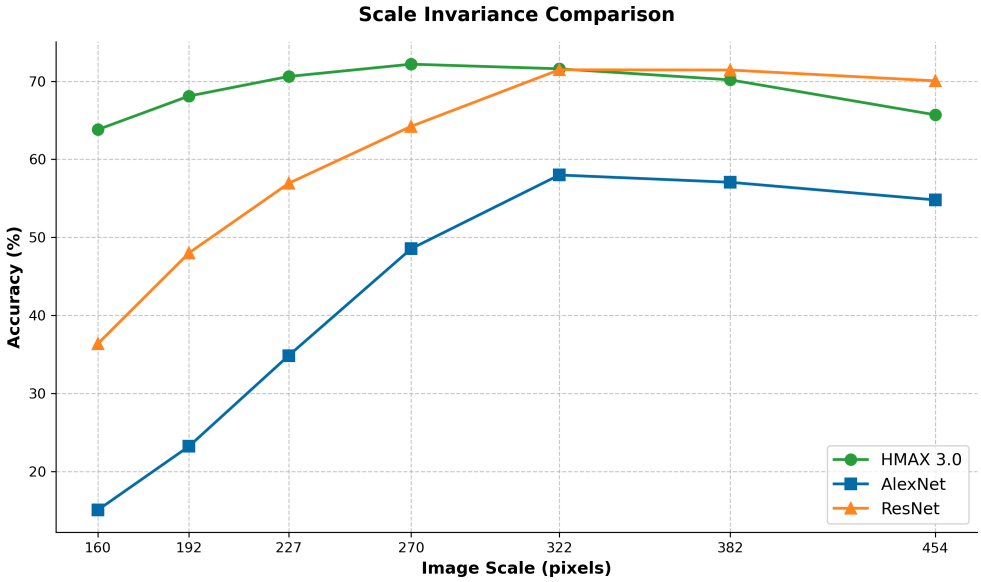

Fig. S3 ImageNet scale invariance evaluation across seven input scales (ported from student draft).